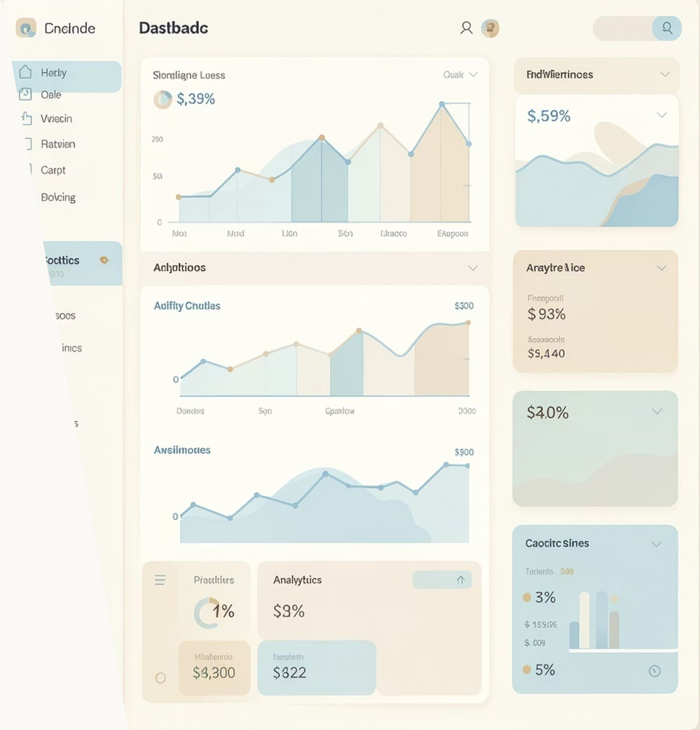
E-Commerce Profitability & Revenue Optimization

Funnel, Unit Economics & Retention Analysis

SQL · PYTHON · POWER BI

~11K ORDERS · ~10K CUSTOMERS



BUSINESS CONTEXT

Growth Without Profitability

- Growth in traffic but no profitability
- High CAC with low returns
- Weak retention limiting LTV

High Acquisition, Low Returns

Marketing spend not generating proportional profit due to elevated CAC.

Poor Conversion Efficiency

Revenue lost at multiple funnel stages before purchase completion.

Weak Retention

Majority of customers are one-time buyers, limiting long-term LTV.

PROBLEM STATEMENT

The Core Business Problem

> Despite strong traffic and revenue, the business is **not profitable** due to inefficiencies in conversion, acquisition cost, and retention.

This is a unit economics problem — not a traffic problem.

Key KPIs Tracked

- Conversion Rate & Drop-off %
- Profit per Order
- Repeat Customer % & LTV

Hypotheses Tested

- Checkout stage has highest friction
- Paid channels may be profit-negative
- Customer base dominated by low-value one-time users

ANALYTICAL APPROACH

Tools & Methodology

SQL — BigQuery

Funnel stage counts, conversion metrics, drop-off %, revenue aggregation

Python — Pandas

Profit per order, loss-making order identification, LTV, customer segmentation, channel profitability

Power BI

3 dashboards: Overview, Funnel & Leakage, Retention & LTV

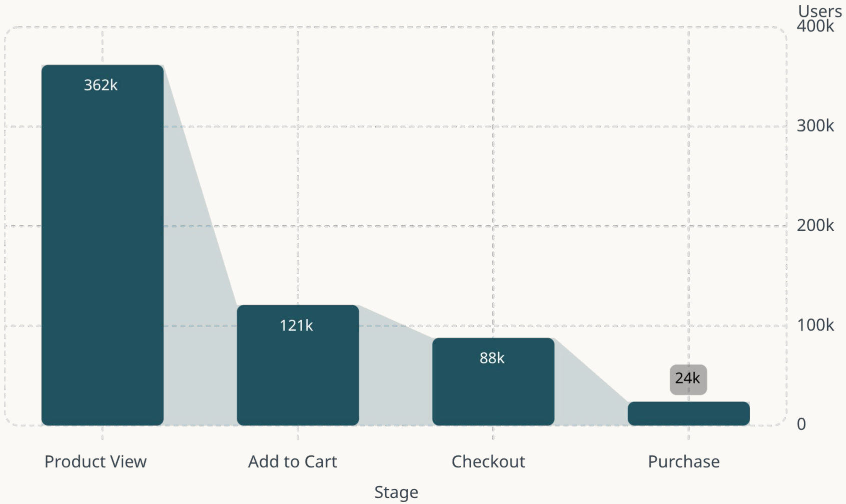
Data: ~11K orders · ~10K customers · Order-level + user-level granularity · Funnel events, revenue, CAC, channel attribution

Analysis performed at order-level and user-level granularity to capture full customer lifecycle.

FUNNEL ANALYSIS

Where Customers Drop Off

6.63%
Total conversion rate



Conversion Rates

View → Cart: 40%

~60% drop — the largest absolute volume loss. Product appeal, pricing, or UX issue.

Cart → Checkout: 46%

~27% drop. Moderate friction at cart stage.

Checkout → Purchase: 55%

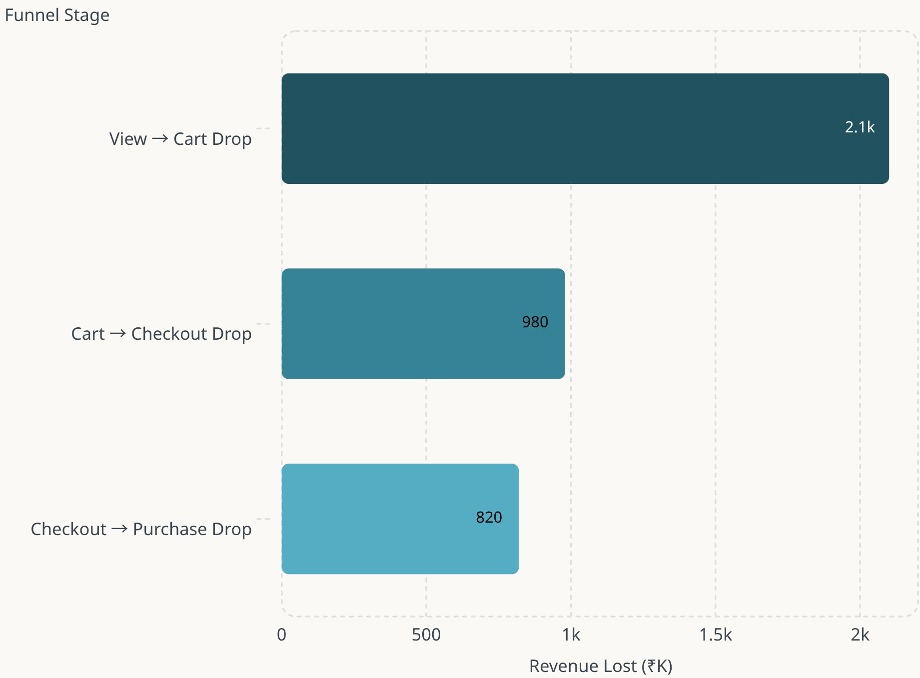
~45% drop. High percentage drop from smaller base, but smaller absolute volume loss than View→Cart.

Largest volume loss occurs at the top of the funnel.

⚠ Overall conversion: **6.7%** — 362K views yield only ~24K purchases.

REVENUE LEAKAGE

₹3.9M Lost to Incomplete Conversions



Key Insight

Revenue leakage (₹3.9M) is more than 2.5× current revenue (₹1.54M).

The early funnel (View → Cart) accounts for the largest share of revenue leakage — not checkout.

₹3.9M+ in potential revenue is lost before a single payment is attempted. This is a direct, recoverable opportunity.

Business Meaning

Fixing product-page conversion has higher ROI than optimizing checkout alone.

PROFITABILITY ANALYSIS

Scaling Revenue, Not Profit

₹1.54M

Total Revenue

~11.5K orders · Avg. order value ₹133

67%

Loss-Making Orders

~7,600 of ~11,500 orders generate negative margins after CAC

60%

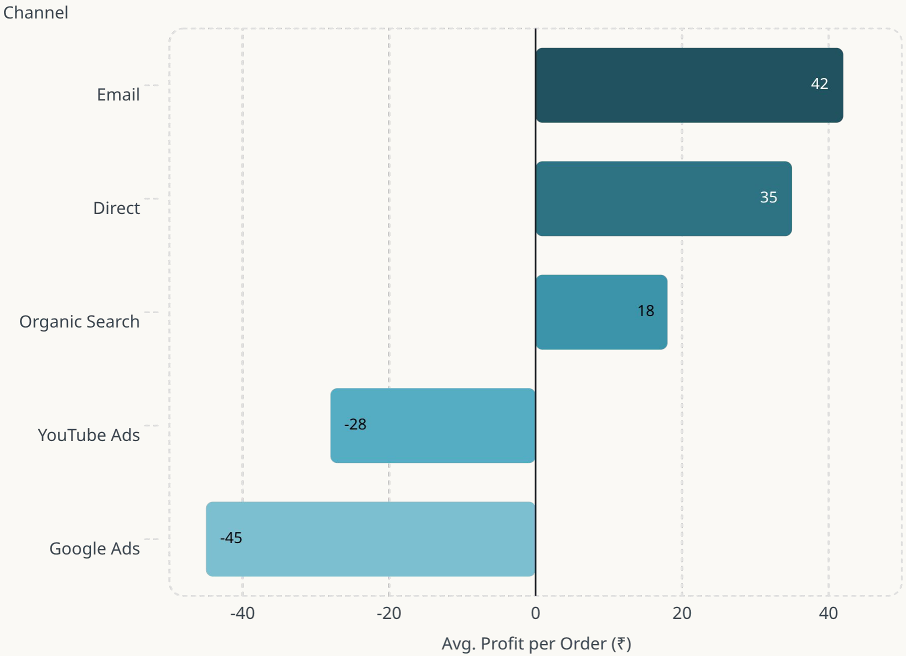
Unprofitable Customers

Only ~40% of customers contribute positive lifetime value

Revenue ≠ Profit. CAC exceeds profit per order for the majority of transactions. Growth is currently scaling losses due to negative unit economics.

CHANNEL PERFORMANCE

Paid Channels Are Profit-Negative



Channel Insight

✅ Direct & Email

Positive returns — low CAC, high intent customers

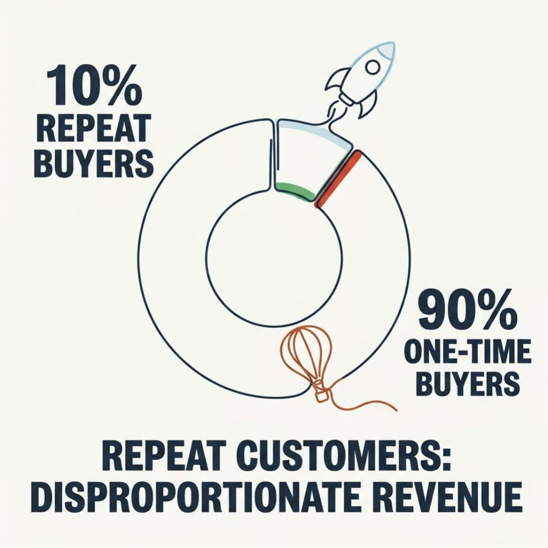
❌ Google & YouTube Ads

Negative profit per order — CAC exceeds revenue contribution

Action: Paid channels drive volume but are currently inefficient — CAC exceeds value. Optimization is required, not elimination. Focus should shift from volume → efficiency.

RETENTION ANALYSIS

90% of Customers Never Return



Retention Findings

90% One-Time Buyers

Customer base dominated by single-purchase users with minimal LTV.

Sharp Retention Decay

Cohort analysis shows retention drops sharply after first purchase — poor post-purchase engagement.

Repeat Segment Drives Value

The small repeat segment contributes disproportionately to total revenue — a high-leverage growth lever.

Repeat Customers (~10%) Drive Value

Repeat customers contribute disproportionately higher value per user.

Retention Is the Highest ROI Lever

Improving retention unlocks the fastest and most efficient growth upside.

Weak lifecycle strategy is the root cause. Acquisition is attracting low-quality, low-intent users.

RECOMMENDATIONS & IMPACT

From Insights to Action

🔴 High Impact

- Improve product page UX to increase View → Cart conversion
- Reduce checkout friction (fewer steps, better payment UX)
- Optimize loss-making paid channels
- Launch retention campaigns (email, loyalty)

🟡 Medium Impact

- Retarget cart abandonment users
- Optimize pricing & discount strategy

🟢 Long-Term

- Personalization & recommendation engine
- Customer lifecycle marketing automation

+10%

Conversion Lift

Projected +8–12% improvement

+15%

Repeat Customers

Through retention campaigns

₹3.9M

Leakage Recovery

Addressable revenue opportunity

Priority sequence: Conversion → Retention → Acquisition